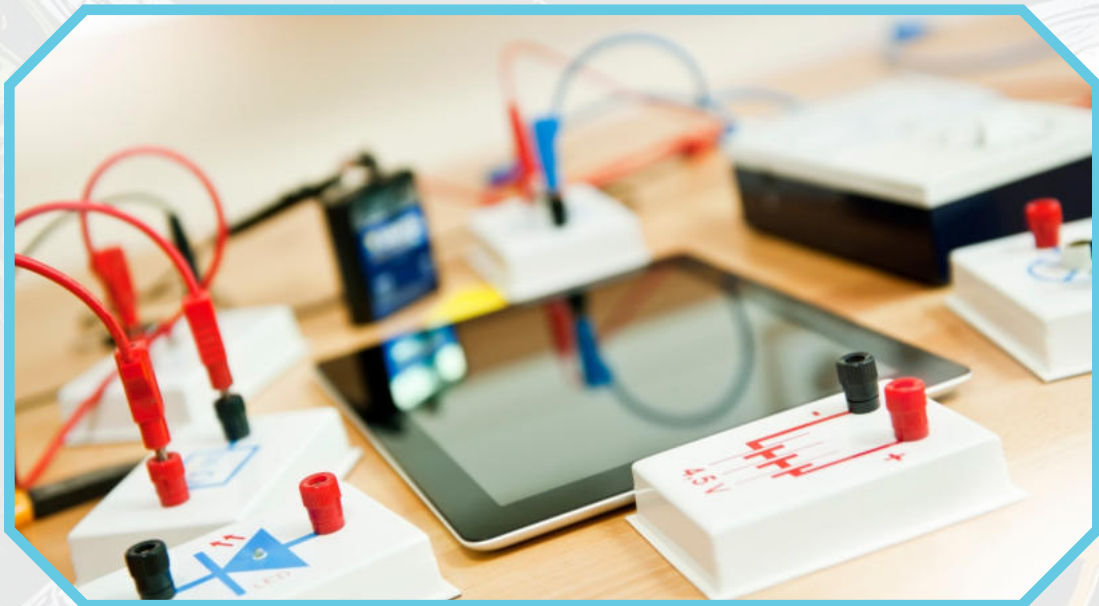




## UNIT 21

# PHYSICS OF SOLIDS



## Classification of Solids:

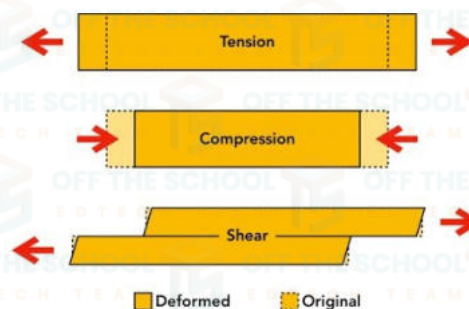
Solids are usually crystalline substances their molecules are arranged in a definite pattern and in fixed positions. The way in which a solid behaves depends on its internal structure and there are three types of solid.

(i) Crystalline solids (ii) Amorphous solids (iii)  
Polymeric solids

## Deformation:

In material science, deformations refer to modifications of the shape or size of an object due to applied forces or a change in temperature.

Deformation is usually caused by forces such as tensile (pulling) or Compressive (pushing)



## Deformation in one dimension (1D):

### Stress:

It is usually defined as force applied to a material per unit area. Stress is a quantity which defines the magnitude of force that cause deformation.

### Strain:

Strain is a measure of how much a material deforms (stretches or compresses) when a force is applied to it.

It is calculated by dividing the change in length by the original length of the material.

## Mechanical Properties of Solids:

Mechanical properties of solids refer to the characteristics that describe how a solid material responds to applied forces or deformation.

Some important mechanical properties of solid are Elasticity, Ductility, Brittleness, and Hardness etc.

## Young's Modulus (modulus of elasticity):

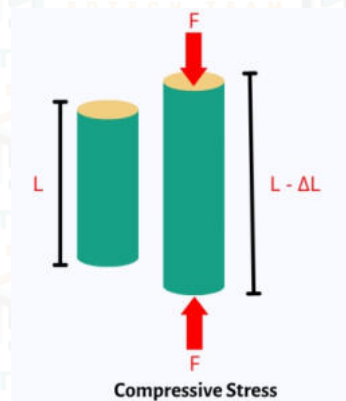
It is a mechanical property of solids that measures the stiffness or elasticity of a solid material and it computes how much a material will deform (stretch or compress) under a given force.

$$Y = \frac{\text{Force}}{\text{Area}}$$



$\gamma$  is denoted by the symbol  $\gamma$  and has (Pascal, pas,  $\text{N/m}^2$ )

Modulus is defined as the ratio of stress to strain.



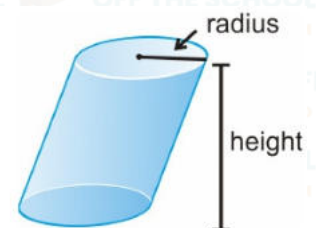
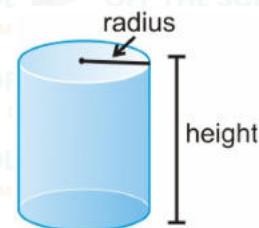
$$\gamma = \frac{\text{Stress}}{\text{strain}}$$

$$\gamma = \frac{\frac{F}{A}}{\frac{\Delta L}{L}}$$

$$\gamma = \frac{FL}{A\Delta L}$$

### Shear Modulus:

Shear modulus, also known as the modulus of rigidity, is a mechanical property that describes a material's resistance to shear deformation when a force is applied perpendicular to the material's surface.



It is denoted by the symbol  $G$  or  $S$  and has units of force per unit area, typically expressed in Pascal (Pa) or Giga Pascal (GPa).

$$G = \frac{F \times L}{A \Delta x}$$

### Bulk Modulus: Compressibility

Bulk modulus, also known as the modulus of compressibility, is a mechanical property that describes the resistance of a material to compression under uniform external pressure. It is denoted by the symbol  $K$  and has units of pressure (usually expressed in Pascal, Pa, or Giga Pascal (GPa)).

### Bulk Modulus:

Bulk modulus is defined as the ratio of the change in pressure applied to a material to the resulting fractional change in volume that occurs in the material. Mathematically, it is expressed as:

$$\frac{\Delta V}{V} = - \frac{p}{B}$$
$$B = - \frac{\Delta P}{\Delta V}$$

Where **B** is the bulk modulus,

**ΔP** is the change in pressure applied to the material,

**ΔV** is the resulting change in volume,

**V** is the original volume of the material.

### Electrical Properties of Solids:

The electrical property of a good conductor depends on its ability to transmit energy without boiling, melting, or changing its composition in any way.

#### Resistivity:

The resistance per unit length and cross-sectional area is called resistivity. It varies from material to material. Its SI unit is Ohm-meter ( $\Omega \cdot m$ ).

#### Conductivity:

It is a measure of a material's ability to conduct electric current, which is influenced by the availability and mobility of charge carriers within the solid.

#### Temperature Coefficient of Resistivity:

The temperature coefficient of resistance is defined as the change in electrical resistance of a substance with respect to per degree rise in temperature. When the temperature increases, the process of electron collision becomes rapid and faster. As a result, the resistance will increase with the rise in temperature of the conductor.

The quantity of charge carriers 'n', the number of charge carriers per unit volume, can be found from the measurement of the Hall Effect. Its SI unit is  $m^{-3}$ .

### Energy Bands in Solids:

When isolated atoms come together to form a solid, interactions between neighboring atoms cause the electron energy levels to split and overlap, creating continuous energy bands.

## Valence Band:

The **valence band** is the range of energy levels occupied by electrons that are bound to atoms and participate in chemical bonding. It is the highest energy band that contains electrons under normal conditions.

## Forbidden Band (Energy Gap):

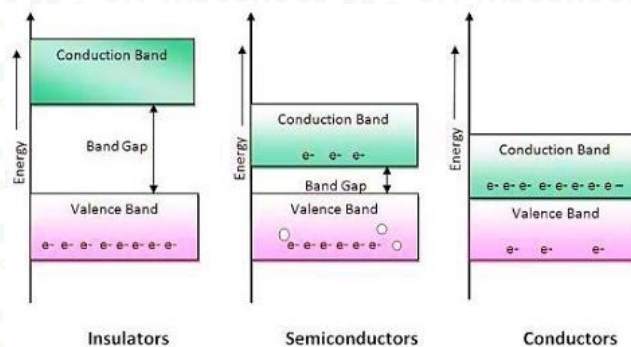
The forbidden band, or energy gap, is the range of energy levels between the valence band and the conduction band where no electron states exist. This gap influences the material's electrical properties; a larger gap typically means the material is less conductive.

## Conduction Band:

The conduction band is the range of energy levels where electrons are free to move throughout the material, allowing electrical conduction. Electrons in the conduction band are not bound to any particular atom and can carry electric current through the solid.

## Classification of Solids:

Depending on the electrical conductivity solids are classified into three main categories: conductors, semiconductors, and insulators.



There are two types of semiconductors:

### a) Intrinsic semiconductor:

When energy is provided to intrinsic semiconductors like silicon and germanium, electrons leave their positions, creating positive holes. Under an electric field, electrons and holes move in opposite directions, enabling electrical conductivity. Group IV elements are considered as intrinsic semiconductors.

### Effect of doping on band structure





### **b) Extrinsic semiconductor:**

Silicon and germanium have low conductivity at room temperature. Adding small amounts of group III or group V impurities, a process called doping, increases their conductivity, resulting in extrinsic semiconductors.

### **Superconductor:**

A superconductor is a material that can conduct electricity without resistance when it is cooled below a certain critical temperature, typically near absolute zero.

### **Curie Point:**

Curie point is also known as Curie temperature. Some magnetic materials lose their magnetic properties as the temperature rises above the Curie temperature. The Curie temperature weakens the magnetic properties of the material.

### **Curie's Law:**

The Curie law states that in a paramagnetic material,

"Magnetization is directly proportional to an applied permanent magnet's magnetic field".

$$M \propto HM$$

$$M \propto xH$$

But when it is heated, the relation is reversed, i.e., the magnetization becomes inversely proportional to temperature.

Mathematically, it is written as:

$$M = C X \frac{H}{T}$$

### **Hard and Soft Ferromagnetic Substances:**

Ferromagnetic materials fall into two main categories: Hard and Soft.

#### **Hard ferromagnetic:**

Materials like neodymium, samarium cobalt, and alnico are permanent magnets with high coercivity, suitable for stable magnetic fields in applications like speakers and generators.

#### **Soft ferromagnetic:**

Including iron, nickel, and cobalt, these materials have low coercivity, making them easy to magnetize and demagnetize.

They are used in devices like transformers inductors that require rapidly changing magnetic fields.